

Cognate redocking as a quality-control step for cryo-EM kinase receptor structures: a study of ATR/CHK1 and CDK-activating kinase

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Received: / Accepted:

Abstract Cognate redocking—reproducing a receptor’s own deposited experimental ligand model—is a standard quality-control step in structure-based virtual screening. Two recent cryo-EM ATR structures (PDB 9L40, 2.87 Å, berzosertib-bound; PDB 9L4B, 3.20 Å, camonsertib-bound) were redocked with AutoDock Vina (2.0 Å symmetry-corrected RMSD pass threshold). 9L4B passed (0.61 ± 0.02 Å), whereas the higher-resolution 9L40 failed (2.53 ± 0.01 Å), not explained by the nearby allosteric pocket (60.65 Å away). The same protocol, extended to 14 cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex (a single-target series), found 13/14 failing. As a pipeline sanity check rather than a comprehensive benchmark, the same pipeline was run on 20 objectively selected CASF-2016 complexes (8/20 passing, 40%; exact 95% CI 19–64%), ruling out a broken pipeline. Neither nominal resolution nor wwPDB map/model metrics (Q-score, residue inclusion) predicted redocking outcome; with only two Q-scored passing structures, this is a descriptive finding, not a definitive statistical correlation. Against a matched rigid control (4.50 ± 0.05 Å), allowing five active-site residues within 4.5 Å of native berzosertib to move improved the 9L40 pose (3.00 ± 0.92 Å) without reaching threshold. Docking a 40-compound pilot library against excluded 9L40 instead of validated 9L4B left the top five ranked compounds unchanged but produced low overall rank concordance (Spearman $\rho = 0.21$) and only 5/10 shared top-10 compounds: receptor choice materially reorders a screen’s mid-ranks while preserving its very top. These results support cognate redocking as a structure-specific quality-control test whose necessity and sufficiency for prospective enrichment require systematic, multi-target evaluation.

1 Introduction

Molecular docking remains one of the most widely used computational methods in structure-based drug discovery, owing to its favorable balance of computational speed and predictive utility [1, 2]. The reliability of docking results depends on a chain of interdependent choices: receptor structure, ligand preparation, scoring function, search algorithm, and validation protocol. Among these, the quality and conformational state of the receptor structure is arguably the most consequential single factor, since errors introduced at this stage propagate through every downstream analysis – a claim this study tests directly rather than merely asserts.

Cryo-EM has emerged as a powerful technique for determining macromolecular structures at resolutions that are increasingly competitive with X-ray crystallography [3, 4]. However, the relationship between nominal resolution and the suitability of a deposited structure for computational docking is not straightforward: even medium-resolution cryo-EM structures, cross-docked under a standard protocol, have been reported to succeed at roughly half the rate of high-resolution crystal structures [5]. Several recent studies have likewise highlighted the importance of per-structure validation in the context of AlphaFold2-predicted models [6, 7], suggesting that structure-specific empirical tests may be necessary regardless of the perceived quality or provenance of the input model.

ATR and CHK1 are established ATP-competitive kinase targets with multiple chemical inhibitors reported for structure-based design [8–13]. Recent cryo-EM structures of the ATR kinase domain—PDB 9L40 (bound to berzosertib, 2.87 Å) and PDB 9L4B (bound to camonsertib, 3.20 Å) [14]—provide substantially improved structural templates for docking compared with the original 4.7 Å ATR–ATRIP complex (PDB 5YZ0). A companion study revealed that berzosertib additionally engages a dimer-interface allosteric site [14], further underscoring the structural complexity of this target system. The practical, protocol-level question this raises is nar-

Key words molecular docking redocking validation cryo-EM ATR kinase virtual screening AutoDock Vina BRICS CASF-2016

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rower than target biology: a rigid-receptor docking run treats a deposited cryo-EM coordinate set as ground truth, so if that single static model does not itself reproduce the experimental ligand pose it was built from, no amount of downstream scoring sophistication can recover a physically meaningful result. We therefore asked whether nominal resolution and standard cryo-EM map-model quality metrics are sufficient to identify structures that can reproduce their deposited experimental ligand poses under a fixed docking protocol.

We note that cognate redocking—reproducing a receptor’s own bound ligand pose—is a necessary sanity check for any docking protocol, but is not itself sufficient to establish prospective screening performance [15]. The present study should be read as a methodological investigation of this validation step rather than as a comprehensive virtual screening campaign. Docking estimates predicted binding and cannot establish functional modulation [8, 9].

2 Materials and methods

2.1 Receptor structures and preparation

Three receptor structures were obtained in mmCIF format from the RCSB PDB AWS Open Data mirror (snapshot 2026-01-01): PDB 9L40 (ATR, bound to berzosertib, 2.87 Å), PDB 9L4B (ATR, bound to camonsertib, 3.20 Å), and PDB 2YM8 (human CHK1, 2.07 Å) [14]. 2YM8 is included as a non-cryo-EM (X-ray crystallographic) kinase reference structure to test whether the observed behavior is specific to cryo-EM-derived receptors; with only one such comparator, this test is necessarily limited to $n = 1$ and is reported as such (Limitations). PDB 5YZ0 (4.7 Å) was excluded as its resolution is insufficient to resolve ATP-pocket side-chain positions reliably [14]. Both protein chains (full homodimer) were retained via Biopython [16].

For 9L40, contact analysis (heavy atoms within 4.5 Å of each protein chain) confirmed residues A2701/B2701 as the ATP-competitive active site and A2702/B2702 as the dimer-interface allosteric site [14]; the 60.65 Å centroid-to-centroid distance between the two is nearly three times the 22 Å docking box used here, ruling out the allosteric site as a confound for the ATP-pocket redocking result.

Receptor preparation was performed with Open Babel 3.1.1 [17] and Meeko (2025). For each structure:

- Hydrogens were added at pH 7.4 (Open Babel -pH 7.4);
- Alternate locations were resolved by retaining the highest-occupancy conformer; no alternate conformers were present among residues within 5 Å of the cognate ligand in any of the three primary structures;
- Crystallographic waters and non-ligand heteroatoms were removed;
- Asp/Glu negative; Lys/Arg positive; histidine tautomer states were assigned by Meeko during flexible-receptor preparation, requiring an explicit HIE assignment for HIS2477 (Section 2.8);
- Meeko preparation automatically deleted ~400 residues whose side-chain density could not be template-matched at approximately 3 Å resolution; automatically dropping surface/loop side chains that cannot be reliably placed in an incomplete cryo-EM map is standard practice for receptor preparation at this resolution regime, not a study-specific deviation, and all deleted residues were >10 Å from the docking box (Supplementary Table S2);
- Proteins were converted to PDBQT with Open Babel using Gasteiger-Marsili partial charges [18] (-gen3d -partialcharge gasteiger).

Physiological pH 7.4 and Gasteiger-Marsili charges were applied uniformly to every receptor and ligand in this study (redocking set, pilot library, CAK survey, and CASF-2016 subset alike) because they are parameter-free, per-atom defaults that require no target- or ligand-specific tuning, making them the appropriate choice for a protocol that is deliberately held constant across a structurally and chemically heterogeneous set of receptors and compounds rather than optimized for any one of them; a more sophisticated charge model or per-compound protonation-state prediction would reintroduce exactly the kind of ligand-specific tuning this study’s design is meant to avoid.

The 22 Å cubic search box was retained as the single, fixed protocol parameter across all receptors and all 45 library/survey ligands (5 seeds, 35 derivatives, and the CASF/CAK cognate ligands), rather than sized per-ligand, because every ligand docked in this study is a single-site, ATP-competitive kinase inhibitor under approximately 550 Da: centered on the cognate ligand centroid, a 22 Å cube comfortably contains such a ligand together with its first-shell pocket residues in every structure tested, so a static box size does not by itself bias the comparison across receptors. This approach follows the recommendation of Feinstein and colleagues [19] that search box dimensions should be justified relative to the specific ligand and receptor rather than chosen by default; a systematic per-ligand box-size sensitivity sweep across the full benchmark was not performed and is noted as a limitation (Section 3.7).

wwPDB validation reports gave per-ligand metrics for the active-site ligand copy used as the redocking reference in each structure: 9L40/A2701, Q-score 0.553, residue inclusion 0.818; 9L4B/A2701, Q-score 0.528, residue inclusion 0.967 [3]. These values indicate both deposited ligand models have comparable local map support, although a comparable Q-score does not by itself establish the correctness of the modeled pose [3].

2.2 Seed library and derivatives

Five compounds formed the seed library: the three deposited active-site ligands (berzosertib, camonsertib, CHK1 inhibitor from 2YM8), plus prexasertib and ceralasertib (SMILES verified against reported molecular formula and exact mass). The seed set was fragmented with the BRICS algorithm [20] into 22 unique fragments, recombined via RDKit [21], capped at 5,000 candidates. Pre-filtering by relaxed Lipinski [22] and Veber [23] rules, plus PAINS [24]/Brenk structural alert removal [25], left 786 candidates. Retained compounds were scored against four drug-likeness rules (Lipinski, Veber, Ghose [26], Egan [27]); 680 of 786 (86.5%) satisfied $\geq 3/4$ rules. The 35 derivatives selected for pilot docking were the highest-ranked by number of rules satisfied, with ties broken by ascending molecular weight. The complete ranking and selection scores are in Supplementary Table S4. This selection protocol is necessarily biased toward smaller, rule-compliant molecules, a limitation that should be considered when interpreting the pilot docking ranking.

2.3 Redocking validation protocol

Each receptor was validated by extracting its deposited ligand, redocking into the identical receptor/box with AutoDock Vina 1.2.7 (Python bindings) [1] at exhaustiveness 32, three independent seeded replicates (seeds 1000, 1001, 1002), and computing RMSD (heavy atoms) against the deposited native pose, with a pre-specified pass threshold of ≤ 2.0 Å [2]. The threshold was chosen to match the Astex Diverse Set convention [2] and was fixed before any receptor was tested, to avoid post-hoc threshold selection.

Two RMSD variants are reported: naive index-matched RMSD and symmetry-corrected RMSD (spyrmsd v0.6.2, graph-isomorphism atom matching [28]). The symmetry-corrected RMSD RMSD_s between a docked pose P_d and the reference pose P_r is defined as:

$$\text{RMSD}_s(P_d, P_r) = \min_{\sigma \in S} \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_{d,i} - \mathbf{x}_{r,\sigma(i)}\|^2}$$

where σ is a permutation of equivalent atoms in the set S of topological symmetries and N is the number of heavy atoms. This correction is important for ligands such as berzosertib that contain internal topological symmetries (equivalent ring atoms, sulfone oxygens), for which index-matched RMSD can overstate the true pose error [28].

Search reproducibility ρ_{repro} was defined as the mean pairwise RMSD between each replicate’s best-scoring pose across the three seeded runs:

$$\rho_{\text{repro}} = \frac{1}{3} \sum_{i < j} \text{RMSD}(P_{\text{best},i}, P_{\text{best},j})$$

This metric characterizes the stochastic variability of the Vina search rather than its accuracy against the native pose, an important distinction this study keeps separate throughout (Results).

Both sanity controls passed: crystal-vs-itself returned exactly 0.000 Å, and an exact 1.0 Å rigid translation was recovered as exactly 1.000 Å. A receptor failing the redocking control in any replicate was excluded from candidate scoring.

2.4 Ligand preparation

For cognate redocking, the deposited ligand chemical state (connectivity, stereochemistry, protonation, tautomer) was preserved unchanged, since the purpose of that test is to ask whether the docking protocol can recover a known experimental pose, not to explore alternative protonation states. For the pilot screen, seed and derivative ligands were 3D-embedded and protonated at pH 7.4 with Open Babel, then assigned Gasteiger-Marsili partial charges before conversion to PDBQT – the same parameter-free protocol used for every receptor and ligand in this study (Section 2.1), rather than a more exhaustive per-compound tautomer/protomer search, which was not run and is not claimed here.

2.5 Pilot library docking

The 5 seeds plus the 35 selected derivatives were docked against each passing receptor using the identical protocol except exhaustiveness 16 (three seeds: 2000, 2001, 2002). Because the protocol consistency argument requires validation at screening-level exhaustiveness, redocking was re-run at exhaustiveness 16: the same pass/fail pattern held (9L40 3.01 ± 0.06 Å fail, 9L4B 0.59 ± 0.02 Å pass, 2YM8 0.92 ± 0.01 Å pass), confirming that the reliability conclusion is robust to this parameter choice.

2.6 Multi-structure survey

The identical validation protocol was applied to 14 additional cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex [29], each bound to a distinct ATP-competitive inhibitor. These 14 structures derive from a single target system and a single research group and therefore represent a clustered, single-target series rather than independent targets. Because the CAK structures are closely related and derive from a single target system, the 13/14 result is treated as descriptive evidence of failure within this series rather than as a population estimate. The pipeline was verified on 9L4B: it reproduced the original result almost exactly (RMSD 0.60 ± 0.03 vs. 0.61 ± 0.02 Å; Q-score 0.528 identical), confirming that the automated survey pipeline is consistent with the hand-curated protocol.

2.7 Positive-control sanity check

Twenty CASF-2016 [30] complexes were selected by an objective, non-random rule fixed before any of the 20 was inspected individually: the first 20 PDB IDs, in file order, of a complete CASF-2016 core-set listing redistributed, not curated or re-ordered by us, in the DeepDock GitHub repository.¹ This sequential selection rule, rather than a stratified or pseudorandom draw, was chosen specifically to avoid any appearance of hand-picking targets favorable to the pipeline. The 285-complex CASF-2016 core set [30] is a widely used benchmark for docking protocol validation, and the 58% Vina self-docking success rate reported for it on this same benchmark [31] is a useful contextual reference. However, this reference value was obtained under specific preparation and sampling conditions different from ours, and therefore should not be interpreted as a universal “Vina success rate.” This control is framed strictly as a pipeline sanity check – confirming the pipeline is not badly broken – and not as a comprehensive docking-accuracy benchmark: 20 complexes from one core set, under one fixed generic protocol, cannot substitute for a systematic accuracy evaluation, and failure to reject a null hypothesis of equivalence to the 58% literature figure does not itself establish that equivalence. Exact binomial 95% confidence intervals are reported using the Clopper-Pearson method [32].

2.8 Flexible-sidechain redocking

As a test for rigid-conformer contributions to the 9L40 failure, five active-site residues within 4.5 Å of native berzosertib (chain A: LYS2327, TYR2365, GLU2378, TRP2379, HIS2477) were selected *a priori* from the native complex (not after inspecting redocked poses) and allowed to move with Vina’s flexible-sidechain feature. The 4.5 Å cutoff is the same heavy-atom contact-shell radius used earlier in this study to disambiguate the ATP-competitive site from the allosteric site (Section 2.1); the resulting five-residue set is therefore a direct, pre-specified consequence of applying that fixed radius to the native pose, not a count chosen for effect.

Rigid and flexible receptor parts were prepared with Meeko (`mk_prepare_receptor.py`), which required an explicit histidine tautomer assignment for HIS2477 (HIE) and, as in Section 2.1, automatically removed template-unmatched surface residues; none of the five target flexible residues were among those removed. The docking box was enlarged from 22 Å (rigid redocking, Section 2.1) to 30 Å to give the flexible side chains room to explore alternate rotamers without a ligand-outside-box error. Because this changes three things relative to the original rigid redocking result at once – side-chain flexibility, box size, and the Meeko-prepared receptor composition

– a matched control was run alongside the flexible run: fully rigid redocking (no residues declared flexible) with the identical enlarged 30 Å box and the identical Meeko-prepared receptor and ligand, isolating the effect of flexibility alone from the other two changes. Both conditions used three seeded replicates (1000, 1001, 1002, exhaustiveness 32, matching the original rigid-redocking protocol). Individual replicate RMSDs are in Supplementary Table S5.

2.9 Rank concordance analysis

The 40-compound pilot set was docked against excluded 9L40 with the same three seeds used for 9L4B (2000, 2001, 2002) for a matched comparison. Rank concordance metrics included: Spearman rank correlation ρ with bootstrap 95% confidence interval (10,000 resamples), Kendall τ , mean absolute rank displacement, and top-10 overlap fraction. The bootstrap procedure resampled compound pairs with replacement to estimate the uncertainty of ρ , accounting for the small set size [33].

2.10 Computational environment and reproducibility

All calculations: Vina 1.2.7 (Python bindings), Open Babel 3.1.1, RDKit 2026.03.6, Biopython 1.88, Meeko (2025), all CPU only on a standard workstation (no GPU acceleration). Exact command-line arguments, prepared PDBQT files, and individual replicate RMSDs are provided in the repository. The three seeded RMSD values for each structure are reported individually in Supplementary Table S6; reported mean $\pm$ SD describes variation among the three docking runs and does not represent inferential uncertainty about receptor suitability. All software versions, random seeds, thresholds, and selection criteria were recorded before the first pipeline execution and are available for independent verification.

3 Results and discussion

3.1 Redocking validation

Table 1 summarizes the redocking validation results. 9L4B and 2YM8 passed with substantial margin (symmetry-corrected RMSD 0.61 ± 0.02 Å and 0.76 ± 0.01 Å). 9L40—the higher-resolution structure—failed (RMSD 2.53 ± 0.01 Å). The allosteric-site confound was ruled out: the nearest allosteric-site ligand copy is 60.65 Å from the active site, nearly three times the 22 Å docking box, so the box could not have reached it. Per the pre-specified rule, 9L40 was excluded from candidate scoring. Figure 1 shows the native and docked poses for both structures.

¹ <https://github.com/OptiMaL-PSE-Lab/DeepDock>, file `Validation_Docking/DockingResults_CASF2016_CoreSet.csv`.

A: 9L40 (berzosertib), FAILED, RMSD 2.53 Å

B: 9L4B (camonsertib), PASSED, RMSD 0.61 Å

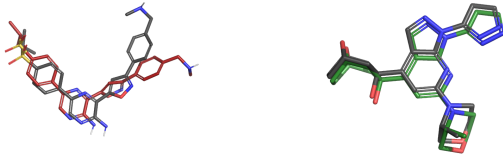


Fig. 1: Native (grey) vs. redocked (colored) best pose. Left: 9L40/berzosertib (symmetry-corrected RMSD 2.53 Å). Right: 9L4B/camonsertib (0.61 Å).

Table 1: Redocking validation (three seeded Vina replicates, exhaustiveness 32)

Structure	Res. (Å)	RMSD (Å)	Pass?	Repro. (Å) ^a
9L40 (ATR)	2.87	2.53±0.01	No	0.56
9L4B (ATR)	3.20	0.61±0.02	Yes	0.045
2YM8 (CHK1) ^b	2.07	0.76±0.01	Yes	0.078

^a Search reproducibility: mean pairwise RMSD between replicate best poses. ^b Non-cryo-EM (X-ray) reference structure.

3.2 Multi-structure survey

Table 2 reports results for all 17 structures tested. Within this sample, 14 of 17 (82%) failed the 2.0 Å threshold; only 9L4B, 2YM8, and one CAK entry (8P75) passed. No statistically significant monotonic association between reported resolution and redocking outcome was detected (Spearman $\rho = -0.28$, $p = 0.28$, $n = 17$). No statistically significant association between the evaluated global measure of residue inclusion and redocking outcome was detected ($r = -0.14$, $p = 0.61$, $n = 16$). Among the 12 structures with a reported Q-score, the two passing structures had a lower mean Q-score (0.569) than the 10 failing structures (0.688); with only $n = 2$ passing, Q-scored structures, this group comparison is a descriptive finding, not a definitive statistical correlation, and should not be read as evidence that Q-score lacks predictive value.

Across the CAK failures, the docked ligand centroid remained close to its native centroid (1.1–4.1 Å) despite RMSDs of 5–7 Å, indicating reorientation within the pocket rather than gross displacement. This pattern is consistent with orientational degeneracy at the kinase hinge – where an ATP-competitive scaffold can occupy more than one plausible binding orientation – or with rigid-receptor limitations that prevent the docking engine from sampling the crystallographically observed binding mode, rather than with a setup error such as a misplaced box.

The 14 CAK structures derive from a single target system and research group and therefore represent clustered observations, not 14 independent targets. Within

Table 2: Multi-structure survey, all 17 structures tested, sorted by symmetry-corrected RMSD. n/a: Q-score not reported for this experimental method (X-ray) or ligand occupancy state (split-altloc CAK ligands).

Structure	Res. (Å)	RMSD (Å)	Pass?	Q-score	Res. incl.
9L4B (ATR)	3.20	0.61	Yes	0.528	0.967
2YM8 (CHK1)	2.07	0.76	Yes	n/a	n/a
8P75 (CAK)	2.00	1.03	Yes	0.609	0.724
9L40 (ATR)	2.87	2.53	No	0.553	0.818
8P72 (CAK)	1.90	2.70	No	0.707	0.963
8P78 (CAK)	1.90	2.98	No	n/a	0.828
8P6Z (CAK)	2.10	4.84	No	0.705	0.885
8P77 (CAK)	1.80	5.02	No	n/a	0.793
8P73 (CAK)	2.00	5.84	No	0.722	0.800
8P6V (CAK)	1.90	6.14	No	0.738	0.966
8P7L (CAK)	2.10	6.16	No	0.708	0.969
8P70 (CAK)	2.00	6.25	No	n/a	1.000
8P76 (CAK)	2.00	6.60	No	0.640	0.735
8P6X (CAK)	1.90	6.60	No	n/a	0.964
8P74 (CAK)	2.20	6.67	No	0.628	0.793
8P71 (CAK)	2.00	7.02	No	0.720	0.926
8P6W (CAK)	1.90	7.33	No	0.762	0.929

this single-target series, 13 of 14 structures failed the pre-specified criterion. Because of this clustering, the 13/14 observation should not be interpreted as an estimate of the underlying failure probability for kinase structures generally; rather, it represents descriptive evidence of failure within this particular CAK series.

3.3 Positive-control sanity check

The unmodified pipeline passed 8 of 20 (40%) CASF-2016 subset complexes selected by the first-20-in-file-order rule described in Methods, an objective and non-random selection fixed before inspection (Table 3). The exact 95% binomial confidence interval (Clopper-Pearson) is 19.1–63.9%. This subset is a pipeline sanity check, not a comprehensive docking-accuracy benchmark, and is read accordingly: because failure to reject a null hypothesis does not establish equivalence, it is treated as evidence the pipeline is not badly broken, not as evidence the pipeline is definitively valid. The observed 40% rate was not statistically distinguishable from the literature Vina value of 58% on the full 285-complex core set [31] (one-sided binomial $p = 0.08$), though this comparison is limited by small sample size and by protocol differences in software version, receptor preparation, and sampling parameters. The 58% reference value represents Vina performance in a specific randomized-ligand redocking experiment and should not be interpreted as a universal “Vina success rate,” as other CASF-2016 evaluations using different protocols have reported substantially higher top-pose success rates (Supplementary Section S1).

The CAK series (1/14 passed, 7.1%; exact 95% CI 0.2–33.9%) showed fewer successful redockings than the

Table 3: Summary of redocking outcomes

Set	<i>n</i>	Pass (rate)	95% CI
CASF-2016 subset (pipeline sanity check)	20	8 (40%)	19.1–63.9%
CAK series (single-target)	14	1 (7.1%)	0.2–33.9%
ATR pair (9L40/9L4B)	2	1 (50%)	1.3–98.7%

CASF sanity-check subset (8/20, 40%). Fisher’s exact test gave $p = 0.037$ for the prespecified one-sided comparison (motivated by the hypothesis that CAK success would be lower than the CASF control) and $p = 0.050$ for the corresponding two-sided comparison. The CASF sanity check did not reveal an obvious gross failure of the docking implementation, whereas 13 of 14 structures in the sampled CAK series failed the prespecified criterion.

3.4 Matched flexible-sidechain analysis

Allowing the five active-site residues nearest the native berzosertib pose to move during docking did not recover it: mean symmetry-corrected RMSD across three seeds was 3.00 ± 0.92 Å (individual seeds: 4.29, 2.23, 2.49 Å), and no seed passed the 2.0 Å threshold. Naively compared against the original fully rigid result (2.53 Å, 22 Å box, Open Babel-prepared ligand), this looks like flexibility made the pose marginally *worse* – but that comparison is confounded, since the flexible run also changes the box size (22 Å $\rightarrow$ 30 Å) and the receptor preparation (Meeko vs. Open Babel) at the same time as flexibility. The matched control isolates flexibility alone: fully rigid redocking with the identical enlarged 30 Å box and the identical Meeko-prepared receptor and ligand, but no residues declared flexible, gives mean symmetry-corrected RMSD 4.50 ± 0.05 Å – *worse* than either the original rigid result or the flexible result. Against this correctly matched baseline, allowing the five residues to move improved the pose from 4.50 Å to 3.00 Å, a real effect in the expected direction; it was simply not enough to reach the 2.0 Å threshold. The larger box size and/or the Meeko receptor-preparation differences, not the choice to keep side chains rigid, account for most of the difference between the original 2.53 Å result and either Meeko-based result. The corrected conclusion is narrower than either the original claim or its opposite: flexibility helps, modestly, exactly as expected if part of the failure is a rigid-conformer effect – but it is not enough on its own to rescue the pose, so a less-supported deposited pose or a conformational change beyond these five side chains remains at least as plausible an explanation (Discussion).

3.5 Error-propagation comparison

Docking the 40-compound pilot set against the excluded 9L40 instead of validated 9L4B produced low rank con-

Table 4: Largest rank shifts between 9L40 and 9L4B scoring

Compound	Rank (9L40)	Rank (9L4B)	Shift
deriv_017	5	34	29
deriv_010	10	39	29
deriv_022	32	7	25
deriv_033	34	9	25

cordance: Spearman $\rho = 0.21$ (bootstrap 95% CI -0.19 to 0.52 , crossing zero), Kendall $\tau = 0.09$, mean absolute rank displacement 11.7 positions (max 29), and only 5 of 10 compounds shared between the two top-10 lists (Table 4). The five shared top-10 compounds were: camonsertib, ceralasertib, deriv_002, deriv_003, and deriv_019 – the same five compounds that occupied the top five ranks in the 9L4B ranking. The top of the ranking is therefore stable across receptors: a campaign selecting only its top five candidates for synthesis would reach the same set either way. The churn is concentrated in the mid-ranks (roughly 6–40), where score gaps are well below Vina’s commonly cited absolute error against experiment (2–3 kcal/mol); a bootstrap ρ CI that crosses zero describes substantial mid-rank reshuffling, not a scrambled ranking overall. The defensible reading is that receptor choice materially reorders the mid-ranks of this pilot screen while leaving its very top unchanged, and that the practical consequence depends on how deep into the ranked list a campaign selects candidates: a top-5 or top-10 selection is comparatively robust here, while a top-20 or top-30 selection would draw on a materially different compound set. This is a demonstration of consequence within this pilot set, not a general enrichment claim, which would require evaluation against known actives and decoys [34].

3.6 Implications for practice

Cognate redocking provides a structure-specific quality-control test that can identify docking failures not apparent from nominal resolution or the evaluated global map/model metrics. 9L40 and 9L4B were solved in the same study, yet only the lower-resolution one passed—an inversion that no proxy metric captured. The observation was reproduced in a separate CAK structure series, although the CAK structures derive from a single target system and therefore do not establish target-independent generality. The 13/14 observed failures in the sampled CAK series should be treated as descriptive evidence within that series rather than as a kinase-wide failure frequency.

No statistically significant association between the available global map/model metrics and redocking outcome was detected in this small survey. Because only two structures with available Q-score measurements passed,

the Q-score group comparison is a descriptive finding, not a definitive statistical correlation, and should not be generalized beyond this sample. Skipping the redocking-validation step this paper argues for is not a neutral shortcut either: the error-propagation comparison above shows that scoring candidates against an unvalidated receptor materially reorders a screen’s mid-ranks even where its very top is preserved, so the practical cost of skipping validation depends on how deep into a ranked list a downstream campaign selects candidates. These results support the use of per-structure cognate redocking as a quality-control step, while systematic multi-target evaluation is required to determine its general utility.

3.7 Limitations

This is a small pilot ($n = 17$). The CAK set is a single-target convenience sample, not a systematic survey of recent kinase depositions. The global Q-score analysis is underpowered (2 passing structures with reported Q-scores). Local density-fit metrics were assessed for 9L40 and 9L4B but a full per-residue active-site analysis across all 17 structures was not performed, as such an analysis would require uniform local validation-report data that were not consistently available for the CAK series. No molecular dynamics, MM-GBSA, or cross-kinase selectivity panel was run. The BRICS derivative library is small (5 seeds, 35 derivatives) and biased toward smaller, rule-compliant molecules; enrichment-based screening metrics cannot be meaningfully computed at this scale. Most importantly, nothing in this study establishes functional modulation—only predicted binding and ranking.

4 Conclusions

Cognate redocking provides a structure-specific quality-control test that can identify failures not apparent from nominal resolution or the evaluated global map/model metrics. Applied to two ATR structures from the same study, the test passed the lower-resolution one and failed the higher-resolution one, an inversion not explained by allosteric-site proximity. Within a single-target CAK structure series, 13 of 14 entries failed the same criterion. Cognate redocking is a useful structure-specific QC test that can expose docking failures not obvious from nominal resolution or the evaluated global map/model metrics, but its necessity and sufficiency for prospective enrichment require systematic evaluation across a broader, multi-target benchmark. Skipping this validation step is not a neutral shortcut: this pilot’s own error-propagation comparison shows that scoring candidates against an unvalidated receptor materially reorders a screen’s mid-ranks, a cost that grows with how deep into the ranking a downstream campaign selects candidates, even in this pilot where the very top of the ranking happened to be preserved. A systematic survey across many independent

targets is the natural next step to determine whether the proportion of failures observed here is representative of recently deposited kinase structures generally. All code, data, and structures are openly available for independent verification.

Acknowledgements

[Placeholder]

Author contributions

CRedit roles (E.Y.: Ethan Ye; A.P.: Aryan Padarthi; S.K.: Sungju Kim; A.E.C.: Alexandre Elie-Dit-Cosaque; E.J.: Eunho Jo). Draft assignment, to be confirmed by all authors prior to submission: Conceptualization – E.Y.; Methodology – E.Y.; Software – E.Y.; Validation – E.Y., A.P., S.K., A.E.C., E.J.; Formal analysis – E.Y.; Investigation – E.Y., A.P., S.K., A.E.C., E.J.; Data curation – E.Y.; Writing – original draft – E.Y.; Writing – review & editing – E.Y., A.P., S.K., A.E.C., E.J.

Data availability

All code, seed and derivative compound tables, redocking validation results, individual replicate RMSDs, prepared PDBQT files, and full residue-deletion lists are available at <https://github.com/ethanyyy123/research-molecule-editing> and in the underlying analysis pipeline repository at <https://github.com/The-MathKing/CogRed>, which additionally documents every review round and correction described in this manuscript. Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8, and the CDK7/cyclin H/MAT1 and CASF-2016 structures cited in Tables 2–3) and were retrieved via the AWS Open Data mirror ([s3://pdxsna.pshots](https://pdxsna.pshots)).

Declarations

Conflict of interest: The authors declare no competing interests.

Ethical approval: Not applicable. This study is a purely computational analysis of publicly deposited structures and did not involve human participants, human data, or animal subjects.

AI assistance: During the preparation of this work, the authors used an AI assistant (Claude, Anthropic) to help develop and debug the computational pipeline’s code, run and log individual analysis steps, and draft and revise sections of this manuscript, and to perform iterative, automated critique of intermediate drafts against the underlying code and data – not a substitute for independent human peer review. That process itself caught

errors, including numbers and described procedures in an earlier draft that were not actually supported by any script or data file in the repository above; those have been corrected or removed. All AI-assisted code, analyses, and text were reviewed by the authors, who take full responsibility for the content of the published article.

References

- Oleg Trott and Arthur J. Olson. Autodock vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 2010. doi: 10.1002/jcc.21334.
- Michael J. Hartshorn, Marcel L. Verdonk, Gianni Chessari, Suzanne C. Brewerton, Wijnand T. M. Mooij, Paul N. Mortenson, and Christopher W. Murray. Diverse, high-quality test set for the validation of protein-ligand docking performance. *Journal of Medicinal Chemistry*, 50(4):726–741, 2007. doi: 10.1021/jm061277y.
- Grigore Pintilie, Kai Zhang, Zhaoming Su, Shiheng Li, Michael F. Schmid, and Wah Chiu. Measurement of atom resolvability in cryo-em maps with q-scores. *Nature Methods*, 17:328–334, 2020. doi: 10.1038/s41592-020-0731-1.
- Werner Kühlbrandt. The resolution revolution. *Science*, 343(6178):1443–1444, 2014. doi: 10.1126/science.1251652.
- Seho Lee et al. Benchmarking applicability of medium-resolution cryo-EM protein structures for structure-based drug design. *Journal of Computational Chemistry*, 44(14):1360–1368, 2023. doi: 10.1002/jcc.27091.
- Matthew Holcomb, Ya-Ting Chang, David S. Goodsell, and Stefano Forli. Evaluation of AlphaFold2 structures as docking targets. *Protein Science*, 32(1):e4530, 2023. doi: 10.1002/pro.4530.
- Valeria Scardino, Juan I. Di Filippo, and Claudio N. Cavasotto. How good are AlphaFold models for docking-based virtual screening? *iScience*, 26(1):105920, 2022. doi: 10.1016/j.isci.2022.105920.
- Akiko Kumagai, Jean Lee, Hae Yong Yoo, and William G. Dunphy. Topbp1 activates the atr-atrtp complex. *Cell*, 2006.
- Trent E. Bass, Jessica W. Luzwick, Gina Kavanaugh, et al. Etaa1 acts at stalled replication forks to maintain genome integrity. *Nature Cell Biology*, 2016.
- Joshua C. Saldivar, David Cortez, and Karlene A. Cimprich. The essential kinase atr: ensuring faithful duplication of a challenging genome. *Nature Reviews Molecular Cell Biology*, 2017.
- Anne M. Casper, Paul Nghiem, Michael F. Arlt, and Thomas W. Glover. Atr regulates fragile site stability. *Cell*, 2002.
- Michelle Debatisse, Benoit Le Tallec, Anne Letessier, Bernard Dutrillaux, and Olivier Brison. Common fragile sites: mechanisms of instability revisited. *Trends in Genetics*, 2012.
- Emmanouil Fokas, Remko Prevo, Jonathan R. Pollard, et al. Targeting atr in vivo using the novel inhibitor ve-822 results in selective sensitization of pancreatic tumors to radiation. *Cell Death & Disease*, 2012.
- Xuejuan Wang et al. Molecular architecture and inhibition mechanism of human atr-atrtp. *Science Bulletin*, 70(13):2137–2146, 2025. doi: 10.1016/j.scib.2025.05.009.
- Ajay N. Jain and Anthony Nicholls. Recommendations for evaluation of computational methods. *Journal of Computer-Aided Molecular Design*, 22:133–139, 2008. doi: 10.1007/s10822-008-9196-5.
- Peter J. A. Cock, Tiago Antao, Jeffrey T. Chang, et al. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 2009.
- Noel M. O’Boyle, Michael Banck, Craig A. James, Chris Morley, Tim Vandermeersch, and Geoffrey R. Hutchison. Open babel: An open chemical toolbox. *Journal of Cheminformatics*, 2011.
- Johann Gasteiger and Mario Marsili. Iterative partial equalization of orbital electronegativity—a rapid access to atomic charges. *Tetrahedron*, 36(22):3219–3228, 1980. doi: 10.1016/0040-4020(80)80168-2.
- Wei P. Feinstein and Michal Brylinski. Calculating an optimal box size for ligand docking and virtual screening against experimental and predicted binding pockets. *Journal of Cheminformatics*, 7:18, 2015. doi: 10.1186/s13321-015-0067-5.
- Jörg Degen, Christoph Wegscheid-Gerlach, Andrea Zaliani, and Matthias Rarey. On the art of compiling and using ‘drug-like’ chemical fragment spaces. *ChemMedChem*, 2008.
- RDKit. Rdkit: Open-source cheminformatics. <https://www.rdkit.org>, 2024.
- Christopher A. Lipinski, Franco Lombardo, Beryl W. Dominy, and Paul J. Feeney. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Advanced Drug Delivery Reviews*, 23:3–25, 1997.
- Daniel F. Veber, Stephen R. Johnson, Hung-Yuan Cheng, et al. Molecular properties that influence the oral bioavailability of drug candidates. *Journal of Medicinal Chemistry*, 2002.
- Jonathan B. Baell and Georgina A. Holloway. New substructure filters for removal of pan assay interference compounds (pains). *Journal of Medicinal Chemistry*, 2010.
- Ruth Brenk, Alessandro Schipani, Daniel James, Agnes Krasowski, Ian H. Gilbert, Julie Frearson, and Paul G. Wyatt. Lessons learnt from assembling screening libraries for drug discovery for neglected diseases. *ChemMedChem*, 3(3):435–444, 2008. doi:

- 10.1002/cmdc.200700139.
26. Arup K. Ghose, Vellarkad N. Viswanadhan, and John J. Wendoloski. A knowledge-based approach in designing combinatorial or medicinal chemistry libraries for drug discovery. *Journal of Combinatorial Chemistry*, 1999.
 27. William J. Egan, Kenneth M. Merz, and John J. Baldwin. Prediction of drug absorption using multivariate statistics. *Journal of Medicinal Chemistry*, 2000.
 28. Rocco Meli and Philip C. Biggin. `spyrmsd`: symmetry-corrected rmsd calculations in python. *Journal of Cheminformatics*, 12:49, 2020. doi: 10.1186/s13321-020-00455-2.
 29. Victoria I. Cushing, Ai Fang Koh, Jing Feng, Kotryna Jurgaityte, Alexander Bondke, Simon H. B. Kroll, Maud Barbazanges, Bernhard Scheiper, Anup K. Bahl, Anthony G. M. Barrett, Simak Ali, Abhay Kotecha, and Basil J. Greber. High-resolution cryo-em of the human cdk-activating kinase for structure-based drug design. *Nature Communications*, 15:2265, 2024. doi: 10.1038/s41467-024-46375-9.
 30. Minyi Su, Qifan Yang, Yu Du, Guoqin Feng, Zhihai Liu, Yan Li, and Renxiao Wang. Comparative assessment of scoring functions: The CASF-2016 update. *Journal of Chemical Information and Modeling*, 59(2):895–913, 2019. doi: 10.1021/acs.jcim.8b00545.
 31. Ute F. Röhrig, Maïte Goullieux, Matthieu Bugnon, and Vincent Zoete. Attracting cavities 2.0: Improving the flexibility and robustness for small-molecule docking. *Journal of Chemical Information and Modeling*, 63(12):3925–3940, 2023. doi: 10.1021/acs.jcim.3c00054.
 32. C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
 33. Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, New York, 1993.
 34. Niu Huang, Brian K. Shoichet, and John J. Irwin. Benchmarking sets for molecular docking. *Journal of Medicinal Chemistry*, 49(23):6789–6801, 2006. doi: 10.1021/jm0608356.